

Part 1: Understanding the data

Que 1.

The unit of observation in the “flights” dataset is an individual flight. Each row in the dataset represents a flight departing from a New York airport in 2013. Variables describe departing time, arrival time, distance, delay, origin, month, day etc for each flight.

```
> colnames(flights)
[1] "year"           "month"          "day"            "dep_time"       "sched_dep_time" "dep_delay"
[7] "arr_time"       "sched_arr_time" "arr_delay"      "carrier"        "flight"         "tailnum"
[13] "origin"        "dest"           "air_time"       "distance"       "hour"           "minute"
[19] "time_hour"
```

There are total 19 variables describing each flight in the data.

Que 2.

Key variables relevant to delay analysis include dep_delay and arr_delay which measure departure and arrival delays in minutes. However, other useful variables are departure time, arrival time, distance, air_time, flight, carrier and origin.

```
> flights[,c("dep_delay", "arr_delay", "dep_time", "arr_time", "distance", "air_time", "flight", "carrier", "origin", "time_hour")]
# A tibble: 336,776 x 10
  dep_delay arr_delay dep_time arr_time distance air_time flight carrier origin time_hour
  <dbl>      <dbl>   <int>   <int>   <dbl>   <dbl>   <int> <chr>   <chr>   <dtm>
1         2         11     517     830    1400     227   1545 UA     EWR   2013-01-01 05:00:00
2         4         20     533     850    1416     227   1714 UA     LGA   2013-01-01 05:00:00
3         2         33     542     923    1089     160   1141 AA     JFK   2013-01-01 05:00:00
4        -1        -18     544    1004    1576     183    725 B6     JFK   2013-01-01 05:00:00
5        -6        -25     554     812     762     116    461 DL     LGA   2013-01-01 06:00:00
6        -4         12     554     740     719     150   1696 UA     EWR   2013-01-01 05:00:00
7        -5         19     555     913    1065     158    507 B6     EWR   2013-01-01 06:00:00
8        -3        -14     557     709     229      53    5708 EV     LGA   2013-01-01 06:00:00
9        -3         -8     557     838     944     140     79 B6     JFK   2013-01-01 06:00:00
10       -2          8     558     753     733     138    301 AA     LGA   2013-01-01 06:00:00
```

This will help us to answer some questions for example-

Which airlines experience highest departure delays?

How does the timing of day affect flight delays? (e.g. morning, evening)

Does distance and airtime affect the flight delay?

Flights arriving from which airport are most likely to delay?

Que 3.

Variables data type:

```
> str(flights)
tibble [336,776 x 19] (S3: tbl_df/tbl/data.frame)
 $ year      : int [1:336776] 2013 2013 2013 2013 2013 2013 2013 2013 2013 2013 ...
 $ month     : int [1:336776] 1 1 1 1 1 1 1 1 1 1 ...
 $ day       : int [1:336776] 1 1 1 1 1 1 1 1 1 1 ...
 $ dep_time  : int [1:336776] 517 533 542 544 554 554 555 557 557 558 ...
 $ sched_dep_time: int [1:336776] 515 529 540 545 600 558 600 600 600 600 ...
 $ dep_delay : num [1:336776] 2 4 2 -1 -6 -4 -5 -3 -3 -2 ...
 $ arr_time  : int [1:336776] 830 850 923 1004 812 740 913 709 838 753 ...
 $ sched_arr_time: int [1:336776] 819 830 850 1022 837 728 854 723 846 745 ...
 $ arr_delay : num [1:336776] 11 20 33 -18 -25 12 19 -14 -8 8 ...
 $ carrier   : chr [1:336776] "UA" "UA" "AA" "B6" ...
 $ flight    : int [1:336776] 1545 1714 1141 725 461 1696 507 5708 79 301 ...
 $ tailnum   : chr [1:336776] "N14228" "N24211" "N619AA" "N804JB" ...
 $ origin    : chr [1:336776] "EWR" "LGA" "JFK" "JFK" ...
 $ dest      : chr [1:336776] "IAH" "IAH" "MIA" "BQN" ...
 $ air_time  : num [1:336776] 227 227 160 183 116 150 158 53 140 138 ...
 $ distance  : num [1:336776] 1400 1416 1089 1576 762 ...
 $ hour      : num [1:336776] 5 5 5 5 6 5 6 6 6 6 ...
 $ minute    : num [1:336776] 15 29 40 45 0 58 0 0 0 0 ...
 $ time_hour : POSIXct[1:336776], format: "2013-01-01 05:00:00" "2013-01-01 05:00:00" "2013-01-01 05:00:00"
```

Total 4 data types are present in the dataset. num represents “Numeric” datatype (including decimals), Int represents “Integer” datatype (whole numbers), chr is for “Character” (category) and POSIXct for date and time.

Missing values: There are total 46595 missing values present in the dataset.

```
...  
> sum(is.na(flights))  
[1] 46595  
> colSums(is.na(flights))  
      year      month      day      dep_time sched_dep_time      dep_delay      arr_time  
      0         0         0         8255         0         8255         8713  
sched_arr_time  arr_delay  carrier  flight      tailnum      origin      dest  
      0         9430         0         0         2512         0         0  
      air_time      distance      hour      minute      time_hour  
      9430         0         0         0         0
```

We can observe that no entire row is empty in the dataset. Majorly the departure - arrival information and airtime values for the flights are missing. We could remove entire empty rows from dataset if present, however, we have few attributes missing and this can be handled by filling the null values by averaging other values of the variable.

Data limitations: We are mostly interested in flight delay analysis. And there could be some other factors affecting delay which needs to be considered. Example- Weather conditions, Season. Additionally, this dataset contains flights data only for 2013. Hence, we cannot consider the conclusions for other years.

Que 4.

As the dataset represents all the flights departing from New York City airports in 2013, it represents population if we're actually interested in only this year and this particular airport. On the other hand, it is only a sample if we consider broader population of worldwide flights. A dataset can be a population or sample depending on the scope of analysis.

Part 2: Numerical Summaries of Delays

Que 5.

Variable: dep_delay

Mean: 12.63 min

Median: -2 min

Standard deviation: 40.21 min

IQR: 16 min

Variable: arr_delay

Mean: 6.89 min

Median: -5 min

Standard deviation: 44.63 min

IQR: 31 min

Que 6.

Delay summary across carrier:

	carrier	avg_dep_delay	avg_arr_delay
	<chr>	<dbl>	<dbl>
1	9E	16.7	7.38
2	AA	8.59	0.364
3	AS	5.80	-9.93
4	B6	13.0	9.46
5	DL	9.26	1.64
6	EV	20.0	15.8
7	F9	20.2	21.9
8	FL	18.7	20.1
9	HA	4.90	-6.92
10	MQ	10.6	10.8
11	OO	12.6	11.9
12	UA	12.1	3.56
13	US	3.78	2.13
14	VX	12.9	1.76
15	WN	17.7	9.65
16	YV	19.0	15.6

Airlines AS and HA show negative average arrival delays, meaning their flights arrive earlier than scheduled on average despite having small positive departure delays. This suggests that these airlines tend to recover time during the flight. AA carrier has average arrival close to zero (0.364 min), indicating that the associated flights arrive almost on time on average. Carriers F9, EV, YV show the highest average departure delays, that means flights operated by these airlines tend to depart significantly later than the scheduled timing. UA and B6 carriers show lower arrival delay compared to departure delay, indicating these flights can cover the distance in air.

Delay summary across origin:

	origin	avg_dep_delay	avg_arr_delay
	<chr>	<dbl>	<dbl>
1	EWR	15.1	9.11
2	JFK	12.1	5.55
3	LGA	10.3	5.78
>			

It is observed that flights departing from EWR airport has highest departure delay as compared to others. Flights from JFK airport tend to make up time in air as they have average departure delay of 12 mins but lower arrival delay (nearly 5 min).

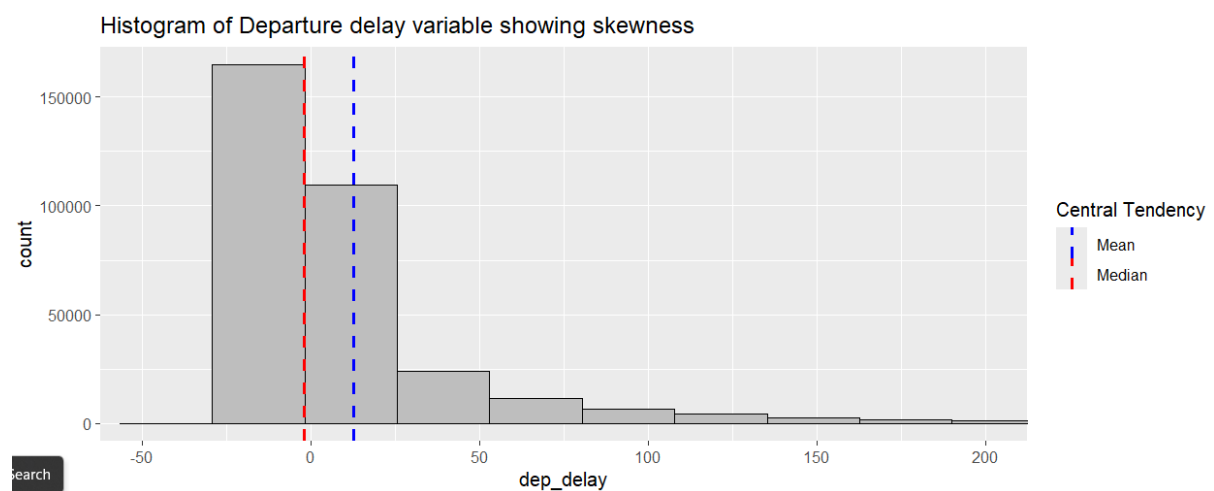
Delay summary across month:

	month	avg_dep_delay	avg_arr_delay
	<int>	<dbl>	<dbl>
1	1	10.0	6.13
2	2	10.8	5.61
3	3	13.2	5.81
4	4	13.9	11.2
5	5	13.0	3.52
6	6	20.8	16.5
7	7	21.7	16.7
8	8	12.6	6.04
9	9	6.72	-4.02
10	10	6.24	-0.167
11	11	5.44	0.461
12	12	16.6	14.9

Overall lower departure and arrival delay is reported for late months of the year – September, October and November. September and October shows negative arrival delay indicating early arrival than the scheduled time. May month shows highest difference in the departure delay and arrival delay, suggesting that flights have covered the highest distance in air to reduce arrival delay in this month.

Que 7.

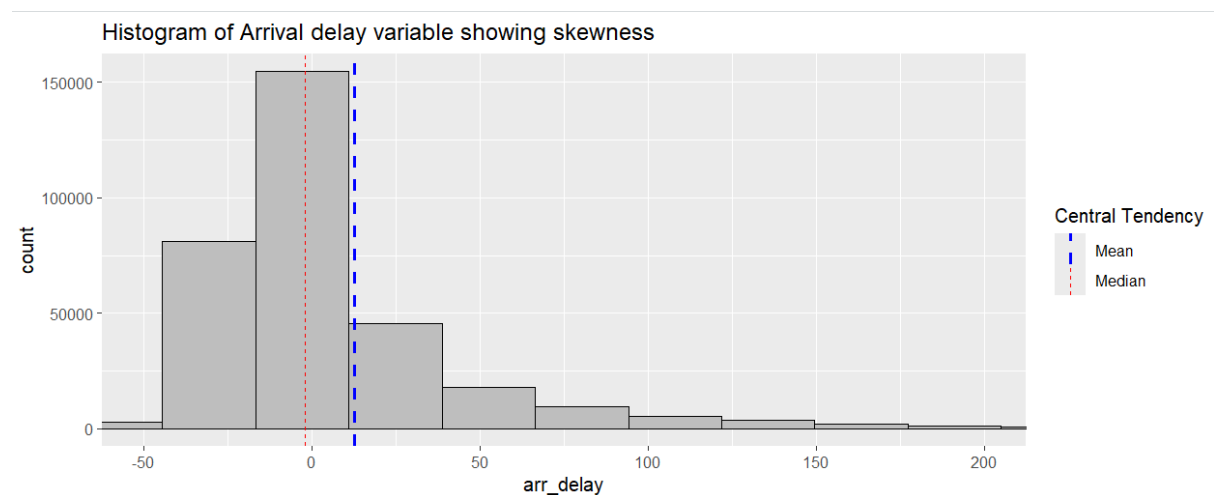
Skewness and differences between mean and median: Departure delay variable



The above plot illustrates the right skewed data or positively skewed data. Here mean value is greater than median. Higher departure delay values have pulled the mean on the right side. This concludes that the number of higher departure delay values are greater than the number of lower departure delay values.

Mean – Median = 14.63 (significant difference)

Skewness and differences between mean and median: Arrival delay variable



The above plot indicates the right skewed data or positively skewed data. Here mean value is greater than median. Higher arrival delay values have pulled the mean on the right side. This concludes that the number of higher arrival delay values are greater than the number of lower arrival delay values.

Mean – Median = 11.89 (significant difference)

Presence of Outliers:

In case of departure delay variable, the 25th percentile (Q1) is -5 and the 75th percentile (Q3) is 11. This indicates that 25% of the flights have delays less than or equal to -5 minutes and 75% of the flights have delays less than or equal to 11 minutes.

The Interquartile range is 16, representing the spread of the middle 50% of the data. Lower and higher threshold values (-29 and 35) were calculated using IQR method. Values below -29 and above 35 are considered outliers.

Total number of lower outliers found to be 4 and the number of higher outliers is 43,212. As high outlier number is greater, this clearly indicate the right skewness of the variable.

In case of arrival delay variable, the 25th percentile (Q1) is -17 and the 75th percentile (Q3) is 14. This indicates that 25% of the flights have delays less than or equal to -17 minutes and 75% of the flights have delays less than or equal to 14 minutes.

The Interquartile range is 31, representing the spread of the middle 50% of the data. Lower and higher threshold values (-63.5 and 60.5) were calculated using IQR method. Values below -63.5 and above 60.5 are considered outliers.

Total number of lower outliers found to be 91 and the number of higher outliers is 27,789. As high outlier number is greater, this clearly indicate the right skewness of the variable.

Mean or Median

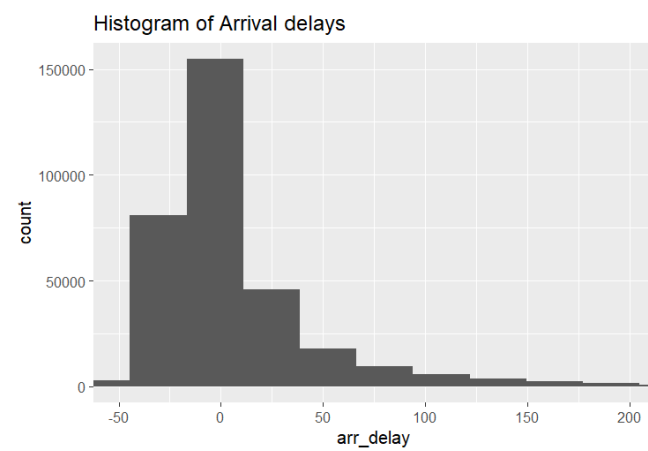
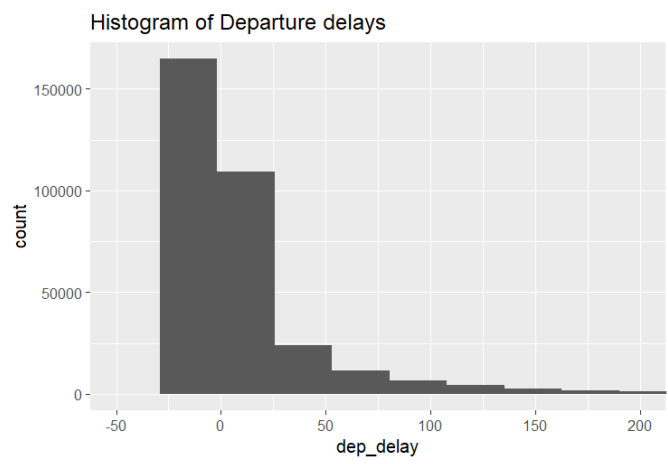
For both delay variables, we observed that the data distribution is positively skewed and mean is greater than median. In such cases, median is considered as more reliable to describe the data because it is less sensitive to the outliers. On the other hand, mean get significantly affected by outliers.

Part 3: Graphical Analysis

Que 8.

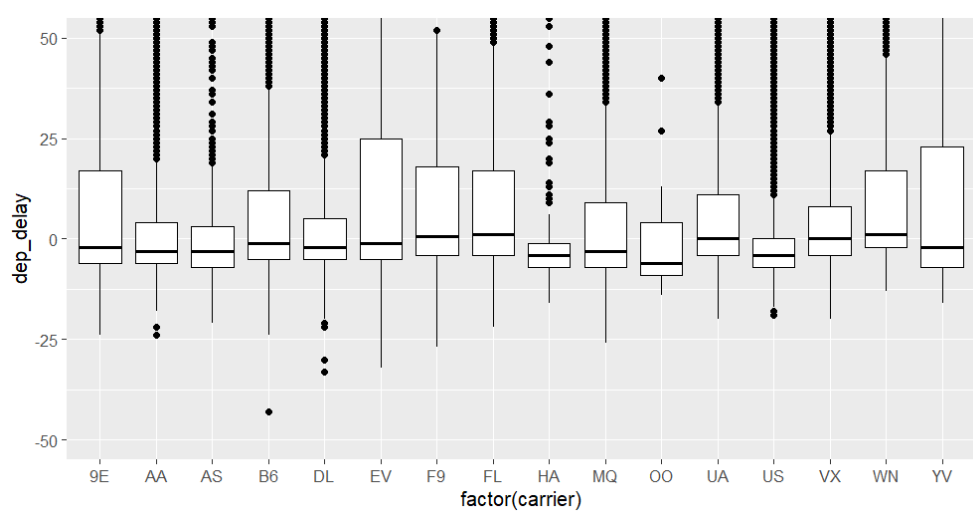
The Histogram of departure delays shows that most flights are concentrated around early departures and small delays. As the delay duration increases, the number of flights decreases significantly, indicating that large delays are rare.

Moreover, the histogram of arrival delays shows that most flights arrived early to their destination. Small number of flights have recorded to be arrived very early (the value goes beyond -50). Similar to departure delay, as arrival delay duration increases, the number of flight decreases significantly.



Que 9.

Departure delay vs Airlines:

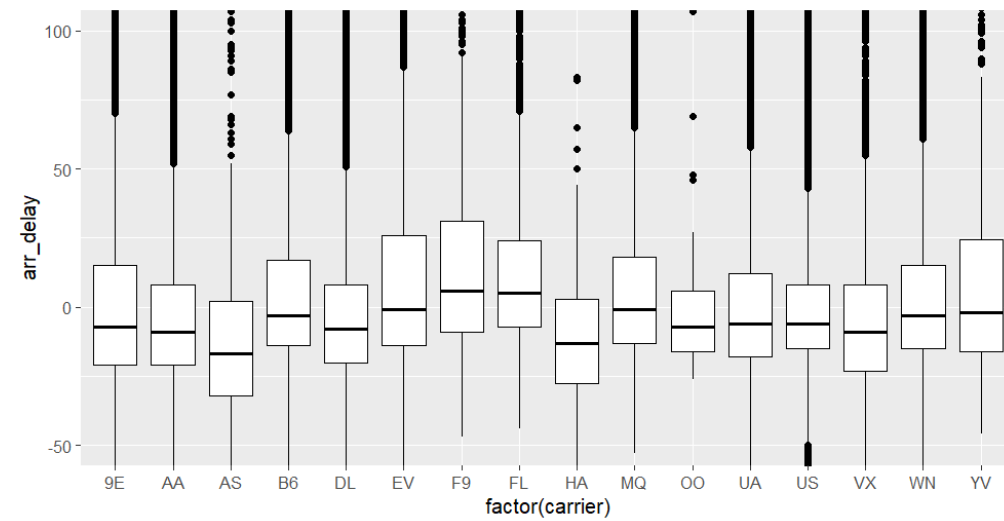


Above boxplots represents that for most airlines, the median departure delay is close to zero. This indicates that flights typically depart on time or with slight delays or early departures. Airlines such as HA and HS exhibit relatively smaller interquartile range this means the spread of (middle 50%)

data is low. Their median values are slightly below zero, suggesting slightly early departure as compared to other airlines.

Airline OO shows significantly low median value, indicating significantly early departures.

Overall, most airlines shows longer upper whisker and larger number of upper outliers, suggesting right skewed distribution. Heavy upper tails have been recorded for airlines such as EV and YV. This indicates that EV and YV airlines flights are extremely delayed.



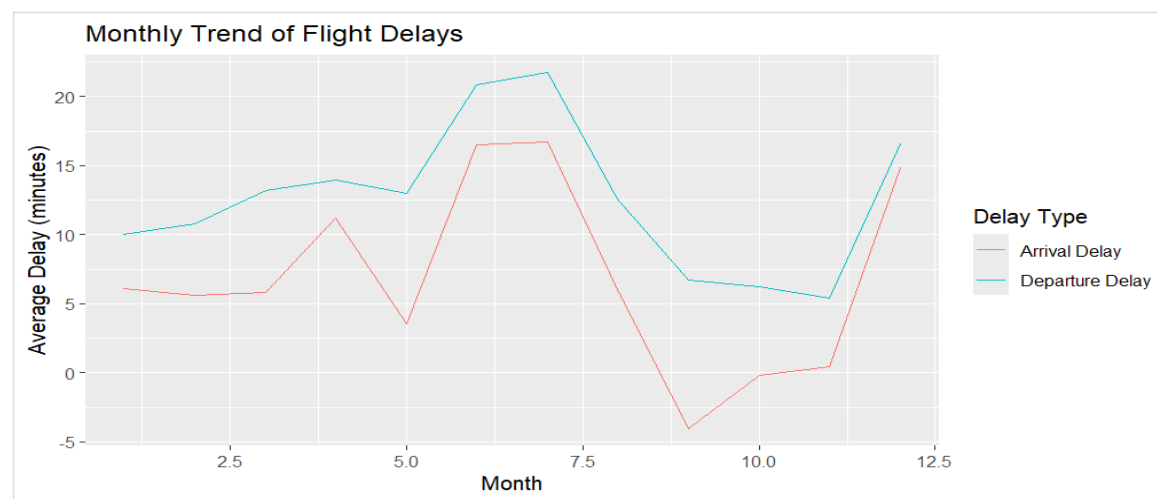
Above boxplots represents that for most airlines, the median arrival delay is less than zero, indicating early arrival.

Airlines AS and HA shows noticeably low median than zero, concluding significantly early arrival of the flight to the destination.

F9 and FL airlines have median greater than zero, which suggests that these flights arrive later to the destination than the scheduled time. Majority of airlines have heavier upper tail representing right skewed distribution.

Que 10.

Monthly trend of flight delays:



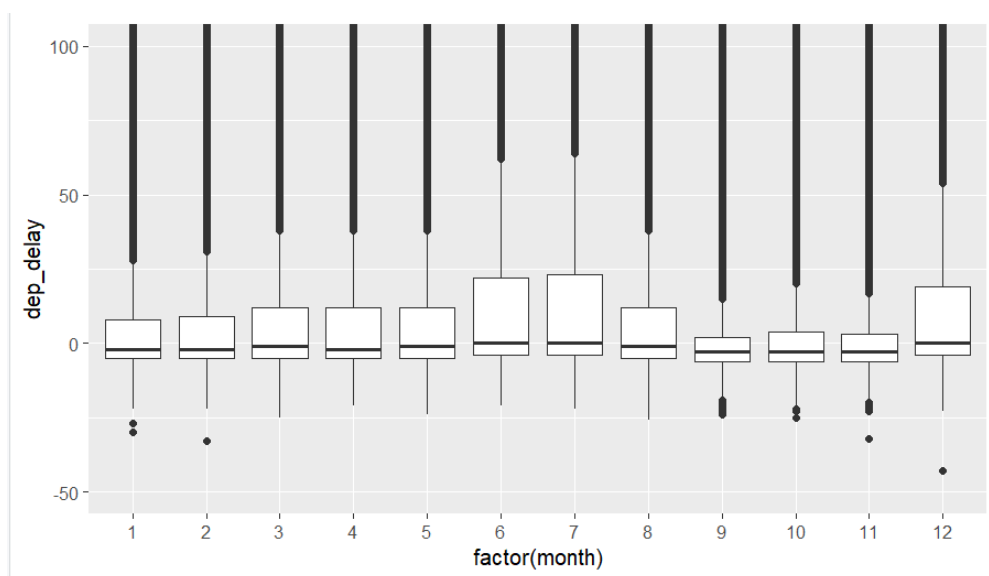
The monthly trend plot shows that both departure and arrival delays rise up steadily from January to July and achieve highest peak in the month of June, before reporting slight reduction in the month in May. Post that, departure and arrival delays sharply decline to (even negative) in the period of September to November. This delay trend found to increase in the last month.

Additionally, it can be observed that departure delays are consistently higher than arrival delays. This indicates that airlines can recover delays during flight. From September to November, arrival delays dropped down beyond zero, suggesting that flights arrive early than the scheduled time in these months are recover delay by making up time in air.

Que 11.

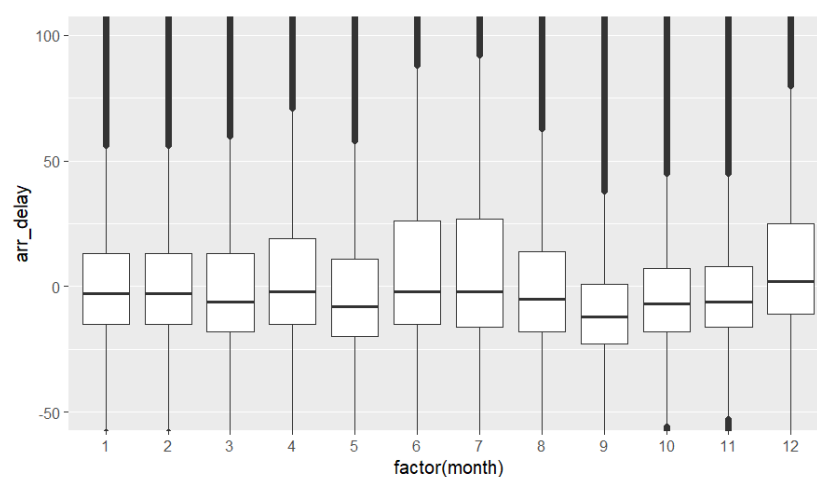
Visualization of choice: Boxplot by Month

Departure delays vs Month



The above boxplot shows that most of the months have departure delay median value nearly zero, indicating on time departure. However, all months have longer upper whisker representing right skewed distribution. Among all months, September, October and November shows short Interquartile range that is low variability in departure delay.

Arrival delays vs Month

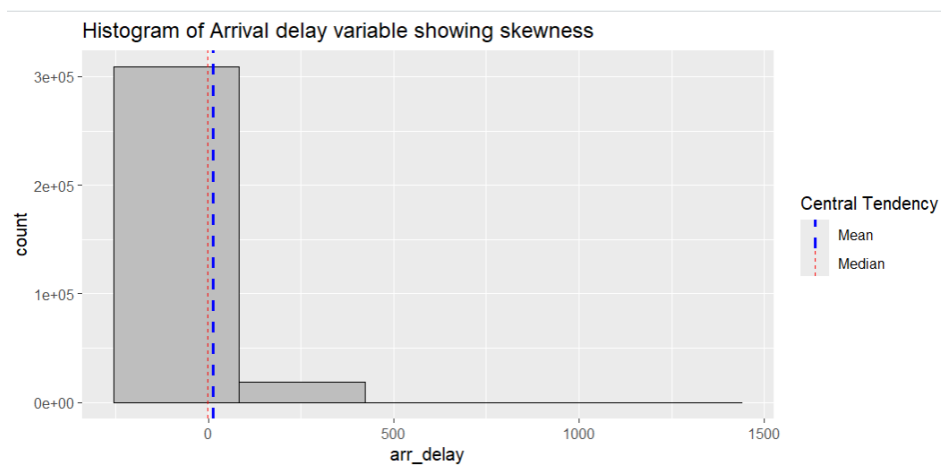
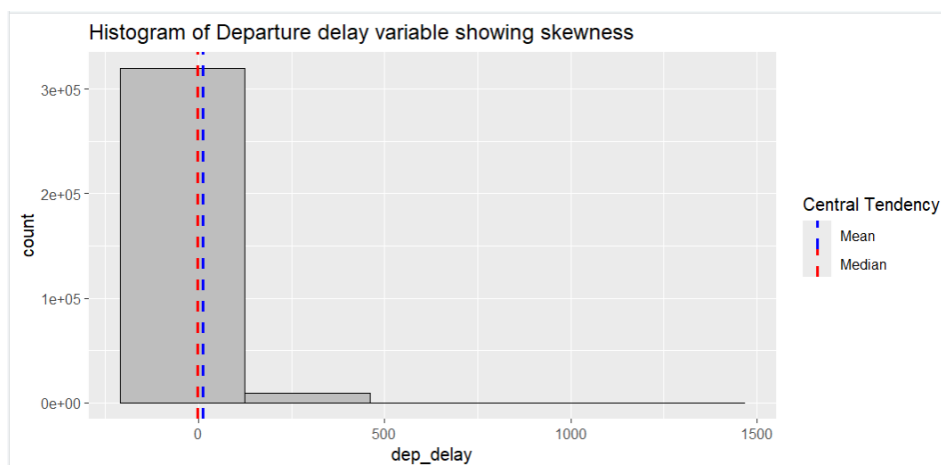


The above boxplot shows that most of the months have arrival delay median value nearly zero. The month of September has significantly lower median than other months, suggesting significantly early arrival on the destination. Similar to the departure delays, arrival delay plot also records heavy upper whisker representing positively skewed distribution.

Que 12.

Misleading data:

Below are departure delay and arrival delay histogram showing skewness. We have used similar code as above question no 7, except providing customized co-ordinates. Additionally, number of bins were also reduced. These changes leads to below plot which fails to display significant difference in mean and median. These plots also hides the trends of complete delay data. While interpreting such misleading plot, one can easily draw such conclusion that mean and median is nearly equal which is completely wrong. This example shows how important is the axis limit and bin number.



Part 4: Distribution and Standardization

Que 13.

The delay distributions are positively skewed and do not have bell shape curve, instead both departure and arrival delay distributions have longer right tail representing unsymmetrical data distribution and large number of outliers on higher value side. To have Normal distribution, the data should be symmetrically distributed and must have bell shape curve. Additionally, the delay distributions do not satisfy normal distribution rule that says Mean should be equal Median. From above answers, we found that there is noticeable difference in mean and median of delays. Because of larger number of higher value outliers, mean is greater than median. Hence, we can safely conclude that delay variables do not show Normal distribution.

Que 14.

The Z score has been calculated for all delay values using below method.

#Que 14.

#z scores of Departure delays

```
mean_delay <- mean(flights$dep_delay, na.rm = TRUE)
```

```
sd_delay <- sd(flights$dep_delay, na.rm = TRUE)
```

```
Z_score_dep_delay <- (flights$dep_delay - mean_delay) / sd_delay
```

```
Z_score_dep_delay
```

#z scores of Arrival delays

```
mean_delay <- mean(flights$arr_delay, na.rm = TRUE)
```

```
sd_delay <- sd(flights$arr_delay, na.rm = TRUE)
```

```
Z_score_arr_delay <- (flights$arr_delay - mean_delay) / sd_delay
```

```
Z_score_arr_delay
```

These scores indicate that how far are those values from the mean in terms of standard deviations.

Que 15.

#Que 15.

#For Departure delay

```
Mean = mean(flights$dep_delay, na.rm = TRUE)
```

```
Standard_deviation = sd(flights$dep_delay, na.rm = TRUE)
```

```
Threshold_dep_delay = Mean + 2*Standard_deviation
```

```
Threshold_dep_delay
```

```
sum(flights$dep_delay > Threshold_dep_delay, na.rm = TRUE)
```

#For Arrival delay

```
Mean = mean(flights$arr_delay, na.rm = TRUE)
```

```
Standard_deviation = sd(flights$arr_delay, na.rm = TRUE)
```

```
Threshold_arr_delay = Mean + 2*Standard_deviation
```

```
Threshold_arr_delay
```

```
sum(flights$arr_delay > Threshold_arr_delay, na.rm = TRUE)
```

2 standard deviation delay refers to the values lie above the product of two standard deviations and mean.

Using R code we have calculated the threshold value (Threshold value = mean + 2*sd) for departure delay – 93.05 and the number of values fall above this threshold is 15,012 which is a significantly large number. This indicates the presence of heavy right tail and confirms that the departure delay variable is positively skewed.

Similarly, we have calculated the threshold value for arrival delay – 96.16. Totally 14,916 values falls above this calculated threshold. This is also a significantly greater number suggesting that arrival delay variable also has right skewed distribution and longer right tail.

Que 16.

In this case, the normality assumption is not reasonable. Both departure and arrival delay data is positively skewed, and mean is greater than median. There are large number of extreme outliers on high value (right side) side. The histogram, box plots clearly indicate a long right tail and this fails to create a bell shape curve or symmetry which are assumptions for normal distribution.

Que 17.

Introduction:

This above analysis describes the patterns in the “flights” dataset. This dataset consists of information on flights departing from New York City airports in 2013, including departure delays, arrival delays, airlines, destinations, distances, time etc. Each row in the dataset represents information about one flight. The analysis mostly focuses on delay variables – Departure delay and Arrival delay. The objective of this analysis is to study major delay patterns and compare these variables with other attributes. We’ve used different descriptive statistical methods for the analysis.

Both Departure and Arrival delay data is positively skewed distributed. From the skewness or histogram plot we observed that most of the flights are on time departed and arrived. Even some departed or arrived early than the scheduled timing. However, small number of flights have shown extreme delay. This gave rise to extreme outliers on right side leading to positively skewed data distribution. Moreover, skewness plot indicates that mean is greater than median in case of both variables.

Statistics:

From the statistical methods, the average departure delay of flights found to be 12.63 minutes, and the average arrival delay of flights found to be 6.89 minutes. The interquartile range has been calculated for both variables.

Additionally, it is observed that the large number of values fall beyond this range, confirming the presence of extreme positive outliers. For departure delay variable, the Interquartile range is found to be 16. Lower and higher threshold values (-29 and 35) were calculated using IQR method. Total number of lower outliers found to be 4 and the number of higher outliers is 43,212. This greater number of higher outliers proves the right skewness. Similarly, for arrival delay variable, the

Interquartile range observed to be 31. Lower and higher threshold values (-63.5 and 60.5) were calculated using IQR method. Total number of lower outliers found to be 91 and the number of higher outliers is 27,789. Again the larger number of higher outliers justifies positive skewness.

Comparisons:

We compared delay data across various variables such as Airlines(carrier), Origin, and Month.

Departure delay and arrival delay variables are plotted against airlines using boxplot. Most of the airlines shows on time departure and arrival. However, some indicates early arrival or departure (AS and HA). Extreme outliers are due to few airlines such as F9, EV, YV which shows highest average departure delays. Few airlines such as UA and B6 show low arrival delay compared to departure delay, indicating these flights tend to recover the distance during flight.

Monthly variation in departure delay and arrival delay has been studied. It is observed that peak delays happened in mid-year. Highest delays are recorded for June and July. On the other hand, this delay value significantly dropped in the months of September, October and November. Even, early arrival is noted for September. Through out the year, departure delays are consistently higher than arrival delay, indicating that flights cover at least some part of departure delay in air, which results in less arrival delay.

Another compared variable is origin. It is observed that flights departing from EWR airport has highest departure delay as compared to others. Flights from JFK airport tend to make up time in air as they have average departure delay of 12 mins but lower arrival delay (nearly 5 min).

Limitations:

The dataset lacks few important attributes- example whether conditions: This will help us to understand if delays are dependent on whether conditions. As per our analysis, most delays happened in month of June and July which are rainy days. We can say that delays depend on seasons, however, whether condition or season attribute would have added evidence for this.

The dataset should also include airport related problems. We've already seen that flights from EWR origin have highest departure delays this point out towards some airport related issues.

We've analysed the patterns of delays but the causes of it is unknown, or we can say the causes can not be justified using this dataset.

Our focus variables (Departure delay and Arrival delay) show right skewed distribution. And it can be viewed using histogram. However, the biggest limitation of the plot is proper number of bins and axis coordinates. If proper number of bins and coordinates are not chosen, plot may show nearly equal mean and median which is misleading. This will lead to wrong interpretations.

Additionally, this dataset only includes flight data of one year which is not sufficient to use these conclusions while predicting future delays.

Insufficient Descriptive statistics:

The delay data is positively skewed and high number of outliers. In such case mean is not a correct representative of data.

In order to use data for future predictions, predictive modelling is required. This can not be achieved by sole descriptive statistics.

Further Analysis:

We can perform Time series analysis on this data. This will help us to know the seasonal trends and how seasons are affecting delays can be studied.

We can remove outliers to reduce the skewness and move the data towards normality. Further, we can merge some another year's data and perform predictive analysis. This will overcome its limitation and resulting conclusions can be used in forecasting.